

Chapter 6

Ethics of Artificial Intelligence and Robotics

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Abstract

This chapter examines the world of ethics and technology and their relationship to Artificial Intelligence (AI) and robots. Incorporating evidence and references with general information and the daily development, this study demonstrates the importance of synchronising AI with ethics, as it discusses the nature of industrialisation and legal responsibilities as a result of manufacturing robots. In the moments of tremendous development, progress and development that artificial intelligence, robotics, learning and automation have encountered, it is rapidly transforming industries and societies all over the world. Reality refers to changing our ways of life, our way of living and our interaction with the external community, all human beings, and even outer space, and this may have been unprecedented for mankind in light of the industrial revolutions because each new industrial revolution may work to develop and improve society, and also these revolutions have the ability on causing major disturbances in our lives and societal norms. To understand modern technologies and avoid any negative impact that may result from them, human society needs to be proactive in managing the industrial revolution without being content with reaction. Our research deals with issues of ethics in the science of AI and robotics and identifying the achievements and strategies of the leading and developed countries in this science, as well as increasing societal awareness and proposing the benefit to all nations.

Keywords: Artificial Intelligence; robotics; robot; ethics; technology

1. Introduction

The development of the steam engine, which paved the way for the shift from agriculture to industrialisation, launched the First Industrial Revolution in 1760. And from this incident, consequently, the switch to the new manufacturing method was made, which led to this shift to the use of coal as the main energy. The Second Industrial Revolution began in 1900, when the internal combustion engine was invented, and this is what is called the era of rapid industrialisation, due to the uses of oil and electricity for huge productivity and providing humanity with what is necessary (Schwab, 2016). In the year 1960 (Nilsson, 1998), they started the Third Industrial Revolution, and it was characterised by the implementation of electronics and advanced technology to automate production. After all these revolutions, the Fourth Industrial Revolution came, which included computers and three-dimensional designs, which could create a solid object through successive layers of materials (Schwab, 2016). On the third digital revolution, which occurred since the middle of the previous century, the Fourth Industrial Revolution is now based.

This revolution is characterised by the merging of technologies in various fields, including physical, digital and biological. From the foregoing, it may be seen that there are three reasons that make the transformations of the Fourth Industrial Revolution more distinctive than that: speed, scope (Schwab, 2016) and the impact of systems. The speed and great breakthrough that took place in the century has no historical precedent at all compared to the revolutions that preceded it. After the occurrence of this Fourth Industrial Revolution, many developments occurred, including at the level of Artificial Intelligence (AI) and robotics, which is an important part of smart digital applications and services. AI refers to the functions carried out by machines that would require intelligence from a human to complete. The world has developed greatly over the past century, and there are now advanced fields that use high-quality technologies such as computers, the internet and what includes the world of technology and development. The development that occurred over the past century was greatly aided by artificial intelligence, which now performs complex tasks and functions that once required human intervention. To maximise the benefits of AI, many businesses are investing heavily. AI is a branch of computer science that seeks to study and design intelligent agents. AI was invented by British mathematician and computer pioneer Alan Turing in 1956. AI has been employed in making and operating robots, where robots follow specific instructions to simulate human behaviour, being faster and more accurate (Wisskirchen et al., 2017). Bots may operate by themselves without human intervention through interactions with websites or having virtual conversations with visitors to websites.

Mostly, bots are useful, but there are those who design them for malicious purposes. The robot is considered a processor, and it can be reprogrammed because it is considered a multifunctional device specially designed to transport materials, parts, tools and specialised devices, all through programming. In general, it is a machine that works instead of humans because it does not get tired and that is because a person may be exposed to injuries and uncomfortable

physical conditions that prevent him from continuing to work and be productive. The issue with this research is that in order to continue and survive, nations must keep up with technological advancements. In order to advance its capacity to create new working methods or create a way to present services for the beneficiaries of the institution or organisation or those dealing with it, the state must allocate a portion of its investments or budgets to spend on AI technologies for the public interest (Wisskirchen et al., 2017).

Ethical policies should be followed in the manufacture of smart machines, such as robots, in a way that benefits humanity, and not in a way that harms humanity, so that it becomes an immoral act. This study focuses on the importance of the worlds of AI and robots in our current era, as robots are saving money by replacing humans at work, and that AI has many contributions in various fields and ethical pronouncement and how AI interacts and robots with ethics and what is the extent of their impact on societies.

2. Literature Review

AI and robotics constitute an elaborate series of research that explores a range of disparate aspects, issues and challenges. One of the most important basic aspects of research in the ethics of AI and robotics is to raise what is important and main in the world of technology and global developments in the modern era (Bostrom, 2014). And this was linked to the legal aspect regarding who bears legal responsibility for the error that results from the robot manufactured by humans, as this raises a question in the world of manufacturing ethics. There is no doubt that there is a great role for different countries in the world of AI and robots.

2.1 AI Concepts, Ethics and Principles

After the entry of AI into our world, it now occupies the jobs of the future that replace the currently known jobs, and AI came with what is known as the 'Fourth Industrial Revolution', so what is AI? What are the ethics of AI?

- *Artificial intelligence*: A subfield of computer science, and the study of smart systems and their design to accommodate the environment and the external environment, where AI is the creation and development of intelligence like human intelligence in terms of the ability to learn, think, analyse, plan, process and then perceive (Abdulzاهر, 2018). AI simulates the method of human intelligence in order to be able to perform tasks instead of humans, although these tasks may include hearing, speaking, movement and a regular manner.
- *Robotics*: The science of robotics is closely related to kinematics and the aspects of mechanical design that it includes, and it has nothing to do with modelling and simulation. Robotics is a new update of AI (Di Gregorio, 2022), where robots are designed by following certain procedures and selecting the best principles to produce what benefits humanity. Robots are useful in various

fields, including engineering fields, as they can work in an environment in which they help humans, greater and greater productivity.

2.2 Characteristics of AI

2.2.1 The Ability to Learn and Perceive

This feature is represented in the ability to understand, learn, and realize human needs so that artificial intelligence devices monitor human behaviour for a person and monitor his routine habits that he follows, such as news that he prefers to see, places that must be visited, or books that he prefers to read ([Hassanin, 2023](#)), so favourite news recommendations and the emergence of advertisements for products. It has not been researched all as a result of the process of learning and realizing artificial intelligence. This is done according to what humans have done before by feeding and indoctrinating the device with some information that makes it capable on understanding and perception, and we conclude from that that these devices do not have the ability to perceive unless they are fed with prior information.

2.2.2 Independence in Decision-Making

This stage is considered one of the most important characteristics and advantages where AI has the ability to infer and conclude, as it is in the stage of understanding and realisation as a result of feeding devices and machines with information, so the step that follows this stage must be represented in making appropriate decisions and quickly choosing a solution or more accurate answer ([Massoud, 2018](#)). We must note that making logical decisions by AI devices is difficult to equate with human inference. Because AI infers after forming a base of knowledge, and then the device or machine has the ability to devise. And the device or machine turns into a semi-human who can learn and understand in order to analyse things and make decisions.

2.2.3 Principles of AI

In order for AI to be properly utilised, we must follow certain principles including ethics, security and humanity:

- *Ethics:* Among the ethics of AI is that it is fair, as it is directed to the age group that benefits from it wherever possible and to avoid bias for a certain group, and AI systems must also be accountable to those in charge of design, development and application, from here we conclude that the system itself cannot be held accountable for its results ([Smart Dubai, 2018](#)). Developers should do their best during development, design and programming, to avoid being held accountable. AI systems must also be transparent. People must be notified whenever AI makes important decisions about them within the limits of

privacy; it should also be as technically explainable as possible, by explaining AI decisions and methodologies to the people that vitally affect them.

- *Security*: AI systems must be safe and subject to human control, so the safety and security of people must be at the top of the priorities when designing any AI system, and the safety and privacy of users must be considered even when these systems are turned off (Smart Dubai, 2018). These systems must not be enabled from causing any harm or sabotage, as it aims to provide services and information, not for revenge and fraud.
- *Humanity*: AI systems must be compatible with human values and contribute to the prosperity of people (Smart Dubai, 2018). As these systems must interact positively, not negatively, with human societies, and their goal is to benefit society, and the concerned authorities must contribute to the development of artificial intelligence and its governance.

2.3 Applications of AI

AI applications offer speech recognition, vision, decision logic and mental simulations, among other capabilities, and it has many applications in various fields to help people in jobs and businesses that may require more effort and more workers' hands, as they can bear trouble and hardship. To reach the highest levels of efficiency, including:

- *Labour and Services*: More than half of the robots available for service in the world, which are more than 7 million robots, are vacuum cleaners, specifically from Roomba 21 (Lund, 2003). There are additional types that perform housekeeping tasks like cleaning floors, ironing clothes, moving objects between rooms and more. Robots are also used in manufacturing, particularly in auto plants, the production of electronics, the food industry and many other sectors.
- *Military and Security*: Robots are used in important matters such as maintaining the security and safety of the country. Some types of robots are designed for espionage and surveillance, whether air, land, underwater or even in space (Lund, 2003); they also help the wounded in wars, search bunkers and attack targets. Some of them are also called (police and security robots). They guard borders and buildings, search for criminals and issue warnings.
- *Entertainment*: One of the main elements in improving watching movies on TV and satellite channels is robots. This is a remarkable and significant development in the world of technology, especially in the field of robotics (Lund, 2003). And one of the most important fields that have been used to achieve great and increasing achievements is the use of robots in films and television sets, and therefore, in this field, it is allowed to depict complex and dangerous scenes using an inanimate entity for greater creativity. Robots have also become a major supporter of festivals and carnivals in order to meet the needs of visitors.

- *Medicine and Healthcare:* Medicine and care have a large share in the field of robotics, and this may benefit the elderly, as it is important to monitor their health conditions, and therefore, the robot can carry out routine activities to help them, such as (eating and drinking) and guiding them through the surroundings (Lund, 2003), and I made some robots to help the medical staff and the patient remotely, while some robots have other advantages because they can perform some of the functions that nurses and pharmacists perform.
- *Deep Learning:* The field of AI has advanced tremendously over the past few years, with a focus on creating artificial neural networks that mimic the way the human brain functions in terms of experimentation, learning and development without the need for human intervention. AI has important aspects in various fields, including car technologies, the traffic aspect, car collisions and predictions to avoid them, clarification of road traffic, knowledge of crowded roads, suggestion of alternative routes and access to places easily (UAE Strategy for Artificial Intelligence, 2017).

The importance may also lie in dealing with human behaviour, and the function of AI is to work on analysing the huge data of individuals and the huge quantities of professional personal information, and through that, human behaviour and interactions can be identified. With the rapid and massive development of AI, it is possible to reach a high rate of accuracy and access to the correct probability of human actions, and from this standpoint, we conclude, as we have already mentioned, that AI is of great importance in many fields such as medicine (UAE Strategy for Artificial Intelligence, 2017), engineering and military and security matters.

The importance of AI is also to alleviate the physical pressures on humans and mitigate risks for them. And these devices have an important role in carrying out the arduous and especially dangerous works that are difficult for a person to perform in relation to drilling and exploration in mines and wells. Robots also have importance in the legal aspect, as it has a major role in arbitration in some issues and assisting the judiciary and the public prosecution in order to achieve justice. Robots may raise the efficiency of lawyers through the application of electronic methods, using the electronic expert system.

In light of this, the government of the United Arab Emirates has committed itself to creating the AI Protocol project, which strengthens efforts to predict the future and adopt its tools in line with the rapid technological transformations that the world is witnessing (UAE Strategy for Artificial Intelligence, 2017). It is positive and supports it in facing the challenges and rapid changes that the world is witnessing and confirms the country's desire to create international alliances to increase the use of the groundbreaking services offered by AI technologies and their contribution to improving human life.

2.4 AI: Examples from Reality

Implementation of AI depends on information and direction given to a such technology to make it effective. AI effectiveness includes improving work quality and reducing cost and complementing human work. Among points to consider are (UAE Strategy for Artificial Intelligence, 2017):

- To make it easier to use applications that rely on this technology, it is important to educate society's members and the public about AI. This will also help government, educational and media institutions work together to create a generation of digital citizens who are able to use these technologies.
- The adoption of this technology in the work and development of services in those agencies will be made easier by raising awareness among institutional leaders, managers and staff of the significance of AI and its uses.
- Enhancing government employees' knowledge of these technologies and developing scientific expertise and specialised local capabilities in the field of AI.
- Launching educational initiatives in universities and schools to keep up with potential future job demands.
- An agreement was reached between the University of Dubai and the Roads and Transport Authority to establish a research centre that supports the transportation industry and advances it using AI in areas like flying taxis, the metro and all other forms of land and water transportation.
- In line with AI, educational institutions are moving towards virtual reality learning technology.
- Giving AI its own course in schools and universities.

2.5 Some Applied Strategies for AI

The UAE AI Strategy, the first significant initiative under the UAE Centennial 2071, was unveiled in October 2017 by His Highness Sheikh Mohammed bin Rashid, Vice President, Prime Minister and Ruler of Dubai. Its goal is to enhance government performance, accelerate achievement and foster innovative and creative work environments with high productivity by investing in the newest technologies and intelligence tools ([UAE Strategy for Artificial Intelligence, 2017](#)). In an optimal manner and in a creative manner that expedites the implementation of development programmes and projects among the main axes of the strategy are the following. Among the main axes of the strategy are the following:

- *Governance*: A law on the safe use of AI, the organisation of a number of conferences to draw in experts in the field ([UAE Strategy for Artificial Intelligence, 2017](#)), the creation of a global protocol with leading governments in the same field, a strong guarantee of governance, effective organisation, the establishment of the State AI Council, task forces of executive heads of innovation in government agencies and more (Bostrom, 2014).
- *Activation*: Organising field visits to government agencies to understand AI and support workshops in all government agencies ([UAE Strategy for Artificial Intelligence, 2017](#)).
- *Capacity Development*: Organising training sessions for government employees in the field of AI, providing the infrastructure and determining a percentage of specialisations for those receiving financial aid from outside the country to study artificial learning are all examples of ways to advance the capabilities of senior government leaders in the field of AI.

- *Application:* The UAE's Ministry of AI made a significant contribution to the creation of tools and applications that organised AI, and in October 2017, it put into practice a strategy that was the first of its kind in history.

2.6 Achievements of the United Arab Emirates in AI

The establishment of the Ministry of AI as part of the state government's new ministerial formation in 2017 signals the beginning of a new phase of development based on AI technologies across all governmental and private economic sectors.

- Executing the government's AI strategy.
- Establishing the UAE as a hub for the creation of AI systems, tools and regulations.

In collaboration with the World Economic Forum, the AI and Robotics Council was established with the goals of setting global standards for robots, developing a global strategy for the use of robots in critical industries like education, health and social services, among others, and providing consultations on the best ways to use robots and AI technologies to improve human life. The 'Emirates Award for Robotics for the Service of Humanity' was introduced as a first step in the initiatives of the AI Council, with the goal of encouraging research and applications of cutting-edge solutions to problems.

Hence, Emirati investment in AI as one of the pillars of the 'Fourth Industrial Revolution', which is considered a major engine for growth and economic diversification, has grown by about 70% over the past three years. One of the most important future directions of the state in the field of AI through the Ministry of Intelligence Synthetic is the direction to activate the followings ([UAE Strategy for Artificial Intelligence, 2017](#)):

- Laying down basic legislation, laws and regulatory frameworks in the country for the governance of AI technologies.
- Promoting and intensifying qualitative investments related to innovation, technological progress and research and development.
- Providing all services through AI and achieving comprehensive integration of AI with medical and security services.
- Employing AI techniques and applications in all economic sectors, including education, health, transportation, roads and others.
- Teaching one million Arab programmers to program AI.
- Strengthening and concerting global efforts and partnerships in the field of AI and building expertise at the international level in cooperation with international companies to attract skills and benefit from the unprecedented services provided by AI technologies and their role in improving human life that transcends geographical borders to include the world.

- Work for the UAE to become the centre of AI in the world by 2030. The economic implications of activating AI technologies at the level of the United Arab Emirates and at the global level.

2.7 AI at National (UAE) and Global Levels

2.7.1 At the Level of the United Arab Emirates

The application of AI techniques has resulted in several things, including reducing costs, changing consumption and production patterns and improving productivity. The results are not limited to that only but extend to achieving high economic growth rates through smart investment in various sectors ([UAE Strategy for Artificial Intelligence, 2017](#)). Several studies have been conducted indicating that AI technologies will be able to stimulate growth in the country's gross domestic product (GDP) by 35% until 2031 and reduce government expenditures by 50% annually. Whether it is related to reducing waste in the number of paper transactions or saving millions of hours that are wasted in completing transactions and the UAE is expected to become the centre of artificial intelligence in the world by 2030. The UAE strategy for AI has achieved annual economic returns in many sectors by raising the productivity of individuals by 13%, avoiding wasting 396 million hours in traditional means of transportation annually ([UAE Strategy for Artificial Intelligence, 2017](#)), reducing transportation costs by 44%, reducing carbon emissions and environmental pollution by 12% and reducing traffic accidents. And the resulting losses by 12% and reducing the need for parking spaces by 20%, and according to a study conducted by McKinsey Corporation, it is expected that 1.9 million jobs will be replaced by other jobs in the Emirates as a result of the application of AI technologies, as well as 65% of the current students in the stages of basic education will not find existing jobs. Today, many traditional skills will disappear in favour of new skills that must be acquired to match the technology of the Fourth Industrial Revolution.

2.7.2 At the Global Level

Investments today in the field of AI globally exceed investments in prospecting for oil. - 47% of current jobs will disappear in the coming years due to artificial intelligence ([Frey, 2017](#)). Automation is expected in the future to replace about 83% of jobs with wages that do not exceed \$20 per hour, which is a matter of concern for its repercussions on unemployment rates and income disparities for countries whose citizens are engaged in these endangered crafts ([Frey, 2017](#)). AI is expected to add more than \$15 trillion in 2030 to the global GDP (10 times global oil sales). The reflection of AI on medical development has added 5 new years to human life in every decade meaning that a person at the age of 90 will be as healthy as a person who is 60 today.

2.8 UNESCO, AI and Ethics

UNESCO has a prominent role in the field of AI, as the conferences held always acknowledge the good and bad consequences, and through it, AI benefits human

societies, ecosystems and human life. There is no doubt that UNESCO, in accordance with the founding provisions and charters, always seeks to maintain peace and security by doing so, through education, culture, communication and information, to strengthen cooperation between nations, in order to ensure comprehensive respect, justice, law and fundamental freedoms for individuals who It is approved by peoples' charters. It takes into account that the means of artificial feeding technology can bring great benefits to humanity and that all countries can benefit from them. UNESCO also acknowledges that AI technologies can exacerbate gaps and disparities within countries at the present time (Anderson, 2007). In general, countries are accelerating in the world of technology, development, use of information and communication technology, AI technology and the need for media knowledge and the digital economy. The means of AI technology benefit the environment and ecosystems, and that reaping these benefits requires not ignoring the damage and bad consequences. Addressing ethical concerns and risks should not hinder innovation and development but rather provide a new opportunity, encourage innovation and link AI technology to fundamental freedoms, ethical values, principles and ethical thinking. UNESCO, in turn, has drafted guidelines on AI and values different systems in the programming and application of AI (vanNorren 2023).

2.9 AI and Robotics Ethics

There may be formation of the concept of 'moral machine'. And it is one of the most important things that raise questions in the world of AI. The fact that these machines are ethical would solve the problems. This sets us up to investigate whether it is possible to create an ethical framework for programming in a machine. The question is that, is it possible to automate ethics? The nature of the robot is devoid of feelings and sensations as it is unable to distinguish between right and wrong (AIHLEG, 2019). Therefore, we cannot attribute the ethics of AI to the robot itself. Rather, the robot maker must follow the manufacturing ethics of this machine as it makes it for the benefit of humanity and not to harm them out of the motive revenge, for example, is immoral, and it should feed him with useful, not harmful information. The bots also lie that they do the same quality consistently (AIHLEG, 2019). This is also an advantage for robots as robots are autonomous, as they will not make bad choices out of anger. When designing robots, complete reliance was on science fiction, as this could be a pain of blessing and another face of curse, so science fiction is a great blessing for them because it gives inspiration and motivation for thinking, experimentation and creativity (AIHLEG, 2019). The expectations of the majority of people from a robot depend more on science fiction than on reality, because reality may lack creativity. This indicates that robot makers may need, during manufacturing, to know that this robot serves humans.

2.10 AI and Robotics and Their Relationship to Law and Ethics

When bot errors occur, it may be unclear who is responsible for any damage caused. Product liability laws are largely untested in the field of robotics, and the

development continues, and therefore, the manufacturers of robots are freed from liability, for example, end-user licence agreements in software, military robotics and supply chain that are only responsible for: the programmer, the manufacturer, Weapons Legal Review Team, Military Procurement Officer, Field Commander, Robot Handler (Calo, 2010). If the robots were more autonomous, it would be reasonable to assign responsibility to the robot itself. Domestic robots are the easiest to manufacture and set up and include home robots that can be monitored or accessed by three parties. Regardless of the manufacture of the robot and its programming, as its use must comply with both law and morality (Calo, 2010). Military robots raise legal and ethical questions, regarding the harm resulting from them when used in times of war. Is this ethical or not?

2.11 Criminal Responsibility for AI Technologies

A person who has full mental strength, and reached the age of maturity and enjoys legal personality, makes a mistake, then he will be responsible for his mistakes, unlike AI and robots (Claussen, 2017). For example, if he made a mistake and his act constituted a violation or a crime that requires responsibility, so how is this done for AI devices such as a robot? There are many crimes committed using AI technologies such as self-driving cars, accidents, home robots, robots and work injuries. The parties who may be criminally liable for this type of crime are the manufacturer, owner or user, the AI technology itself or a third party. However, opinions were divided on the extent to which the rules of criminal responsibility are applied to applications of AI. The first trend tended to apply them without any conditions because they fulfill the material and moral element of the crime. The second trend sees the possibility of applying the criminal liability to artificial intelligence applications, provided that the following conditions are met (Claussen, 2017):

- Equipping AI with algorithms that enable it to make non-trivial ethical decisions.
- The application should be able, in light of the algorithms, to communicate its ethical decisions to humans.
- To allow the application to work in its environment without direct human supervision, as the smart application must make this decision by itself.

There are two areas for imposing responsibility, first when the programmers or users are negligent without intent to commit the crime and the error resulted based on negligence; in this case, there is no reason not to hold them responsible for the crime. Secondly, when the programmers knowingly and deliberately commit the crime via the intelligent entity, for example (Claussen, 2017), if we have an autopilot and it is negligently programmed to defend its mission without any restrictions on the loss of human lives. The second type has to programmer the AI entity to commit a bank robbery (Claussen, 2017). As a result, if the AI entity acts as an innocent agent without knowing anything about the criminal risk, it does not bear criminal

responsibility for the crime it committed. On the other hand, if the AI entity does not act as a mere innocent agent in addition to the criminal liability of the programmer or user according to the natural principle, the AI entity itself bears criminal responsibility for the crime.

2.12 Robots in Future, Especially in Japan

In the future, it is expected that robots will play a fundamental and complex role that has a wide scope in society, especially with the remarkable development of AI, for example: Police robots equipped with biometric decisions and sensors could detect weapons and drugs from faces within a certain distance, and military robots might make attack decisions on their own (Mitchell et al., 2018). Nowadays, most robots have a human behind them, whereas robots are multipurpose and may serve the needs of domestic workers, unlike teams of robots, each of whom has its own way. In some countries, including Japan, robots are considered alternatives to humans, as the elderly population increases, birth rates decrease and thus the labour force shrinks. Robots are designed to fill the workforce gap. Given the Japanese people's love for technology, it is not surprising that there is one among the 25 robot workers. In terms of military robots, the United States dominates, and Japan and South Korea have a share in the field of social robots, such as caring for the elderly.

3. Conclusion

In conclusion, today, the term AI has become one of the terms that people use frequently, especially with the development of smart worlds and modern technologies. Where he provides tasks that may require human intervention, but he is the one who performs them alone. Routine human work can be reduced using AI to same time, and efforts with a high level of accuracy. The latest AI has become a huge revolution in the modern era because most of the data need AI and has become a saviour for companies because of its many benefits. Likewise, robots have become a necessary thing, especially in some industries and very difficult tasks because a person risks his life. Therefore, the invention of a robot is beneficial to humanity if the correct ethics are followed.

If the robot is used properly first because the ethics of its manufacture do not lie with the robot itself but rather attributed to robot manufacturers, ethics may be the basics of building a robot. Theoretically, the use of AI technology enhances the performance and productivity of institutions by automating procedures or tasks that previously required human labour. The theoretical implications are related to the conclusions that have been reached related to ideas and theories in this field. Regarding the applications, modern AI enables us to swiftly handle massive volumes of data. These narratives aid in understanding societal patterns, market trends and climatic changes. Major developments in robotics and AI have been made in the fields of home automation, communications and medicine. The replacement of the judiciary by robots, or the link of AI and robotics to the law in

general, may be covered in detail in the next study. Therefore, a robot manufacturer must follow the instructions and teachings of ethics and law when manufacturing the robot because it is unreasonable to manufacture something by human hands in order to harm the humans themselves.

References

- Abdulzaher, M. (2018). *Artificial intelligence journalism: Fourth industrial revolution and media restructuring*. Dar Badael for Publishing and Distribution.
- AIHLEG. (2019). *High-level expert group on artificial intelligence: Ethics guidelines for trustworthy AI*. European Commission.
- Anderson, S. L. (2007). Machine ethics: Creating an ethical intelligent agent. *AI Magazine*, 28(4), 15–26.
- Calo, R. (2010). Robots and privacy, In P. Lin, G. Bekey, & K. Abney (Eds.), *Robot ethics: The ethical and social implications of robotics*. MIT Press. <https://ssrn.com/abstract=1599189>
- Claussen, M. (2017). *Intelligence and external element of crime. An analysis of the liability problem* (pp. 1–51). Orebro Universitee.
- Di Gregorio, R. (Ed.). (2022) *Kinematics and robot design IV, KaRD2021*. MDPI Books.
- Frey, C. (2017). *Technological forecasting & social change*. University of Oxford IPT.
- Hassanin, M. (2023). *Artificial intelligence and civil liability for applied damaged*. Cairo University.
- Lund, H. H. (2003). Adaptive robotics in the entertainment industry. In *Proceedings 2003 IEEE international symposium on computational intelligence in robotics and automation. Computational intelligence in robotics and automation for the new millennium (Cat. No. 03EX694)* (Vol. 2, pp. 595–602). IEEE.
- Massoud, N. (2018). Responsibility of doing electronic systems. *The Annals of Algeria*, 1(31), pp41.
- Mitchell, T., Cohen, W., Hruschka, E., Talukdar, P., Yang, B., Betteridge, J., Carlson, A., Dalvi, B., Gardner, M., Kisiel, B., Krishnamurthy, J., Lao, N., Mazaitis, K., Mohamed, T., Nakashole, N., Platanios, E., Ritter, A., Samadi, M., Settles, B., ... Welling, J. (2018). Never-ending learning. *Communications of the ACM*, 61(5), 103–115. <https://dl.acm.org/doi/10.1145/3191513>
- Nilsson, J. (1998). Artificial intelligence: A new Sunthesis. <https://doi.org/10.1016/c2009-0-27773-7>
- Schwab, K. (2016). The fourth industrial revolution: What it means and how to respond. <https://www.weforum.org/agenda/2016/01/the-fourth-industrial-revolution-what-it-means-and-how-to-respond/>
- Smart Dubai. (2018). Principles and guidelines of ethics of artificial intelligence. https://www.digitaldubai.ae/pdfviewer/web/viewer.html?file=https://www.digitaldubai.ae/docs/default-source/ai-principles-resources/ai-ethics-ar.pdf?sfvrsn=d4184f8d_6
- UAE Strategy for Artificial Intelligence. (2017). <https://u.ae/en/about-the-uae/strategies-initiatives-and-awards/strategies-plans-and-visions/government-services-and-digital-transformation/uac-strategy-for-artificial-intelligence>

- VanNorren, D. E. (2022). The ethics of artificial intelligence, UNESCO and the African Ubuntu Perspective. *Journal of Information, Communication and Ethics in Society*, 21(1), 112–128. <https://doi.org/10.1108/JICES-04-2022-0037>
- Wisskirchen, G., Biacabe, B. T., Bormann, U., Muntz, A., Niehaus, G., Soler, G. J., & von Brauchitsch, B. (2017). Artificial intelligence and robotics and their impact on the workplace. *IBA Global Employment Institute*, 11(5), 49–67.